

Data analyst with hands-on experience in e-commerce data processing, data quality optimization, SQL-based investigation, rule iteration, sample screening, and workflow tooling. Experienced across data analysis, machine learning, and natural language processing scenarios, with a track record of translating business requirements into reliable data operations and lightweight internal tools.

## • EDUCATION

### The University of Texas at Austin

Aug 2018 - May 2022

Bachelor of Science in Mathematics

GPA: 3.6 / 4.0

## • EXPERIENCE

### NielsenIQ, Shanghai Branch

Dec 2024 - Present

Senior Data Analyst

- Supported multi-dimensional data requests from client-facing teams by extracting, cleaning, and validating large datasets with SQL, enabling configuration impact analysis and before/after comparisons for product classification results.
- Investigated bad cases in a BERT + rules product classification workflow, explained data quality issues to clients, and converted recurring error patterns into SQL / PL-SQL correction rules deployed in HUE and Oracle pipelines.
- Reduced final delivery error rate from approximately 20% to below 10% through iterative issue diagnosis, rule refinement, and structured validation.
- Designed a high-risk sample screening approach using locally deployed BERT vectorization in PyTorch, delivering 140,000+ candidate samples from three years of historical data for manual validation, supplemental labeling, and future model retraining.
- Built a lightweight Streamlit + Oracle productivity tool to streamline Excel ingestion, ad hoc data pulls, configuration lookup, and result comparison workflows.

### Shanghai Xuankai Business Consulting Co., Ltd.

Nov 2022 - Nov 2024

Business Analyst

- Participated in market research projects for luxury goods and sports footwear/apparel brands, covering questionnaire data processing, desk research, and consumer and market information analysis.
- Synthesized research findings into client-facing insights, prepared presentation materials, and supported proposal communication to translate research outputs into business recommendations.

## • PROJECTS

### Multimodal LLM Validation for Product Attribute Recognition

Feb 2026 - Present

- Designed batch concurrent API calling scripts and iterated prompts to evaluate multimodal model feasibility for product attribute recognition where text-only workflows were insufficient.
- Identified generalizable attributes such as specifications and packaging type as stronger candidates for practical implementation and complex sample triage.
- Planned an Agent workflow extension for task decomposition, API orchestration, result checks, and human-review handoffs.

### OCR + Model Number Matching Automation

Aug 2025 - Dec 2025

- Used PaddleOCR to extract product model information for electronics and durable goods catalog maintenance, then combined field cleaning and brand naming rule analysis to improve matching and code-linking efficiency.
- Derived brand/model naming rules for products that could not be directly matched, focusing manual review on out-of-rule samples.

## • SKILLS

#### Data Analysis and Modeling

Python (Pandas, NumPy, Scikit-learn, PyTorch), SQL, R, A/B Testing

#### Data Platforms and Tools

Hadoop ecosystem (HDFS, HUE, SparkSQL), Oracle, MySQL, Power BI, Streamlit

#### AI Tools and LLMs

Cursor, GitHub Copilot, Claude Code; Qwen/Doubao APIs

#### Office and Automation

Microsoft 365 (Excel, PowerPoint, Word), Power Automate